2nd Order DEs: Non-Homogeneous

Questions 1–43

Question 1. Given that $y = Ae^{-2x} + Be^{-3x}$, show that $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$.

a) Show that if $y = k$, where k is a constant, then $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 6k$.

b) Hence find a constant K such that $y = K$ is a solution of the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 3$.

c) Show that $y = Ae^{-2x} + Be^{-3x} + K$ is a solution of the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 3$.

Question 2. Find a particular integral for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 20y = 5$

b) $4\frac{d^2y}{dx^2} - 9y = 6$

c) $4\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 3y = -60$

Question 3. Using your answers to question 2, find the general solution for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 20y = 5$

b) $4\frac{d^2y}{dx^2} - 9y = 6$

c) $4\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 3y = -60$

Question 4. It is given that $\frac{d^2y}{dx^2} - \frac{dy}{dx} = 8$.

a) Show that a particular integral of the form $y = k$, where k is a constant, is unsuitable.

b) Show instead that a particular integral of the form $y = ax + b$, where a and b are constants, is suitable.

c) Hence find the general solution of the differential equation.

Particular Integrals of the form $y = ax + b$ where a and b are constants

Question 5. Find the values of a and b such that $y = ax+b$ is a solution of the differential equation $\frac{d^2y}{dx^2}+3\frac{dy}{dx}-10y = 10x + 7$.

Question 6. Find a particular integral for each of the following differential equations.

a) $\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 15x + 4$

b) $9\frac{d^2y}{dx^2} - y = 5 - x$

c) $4\frac{d^2y}{dx^2} + 16\frac{dy}{dx} - 9y = -45x$

Question 7. Using your answers to question 6, find the general solution for each of the following differential equations.

a) $\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 15x + 4$

b) $9\frac{d^2y}{dx^2} - y = 5 - x$

c) $4\frac{d^2y}{dx^2} + 16\frac{dy}{dx} - 9y = -45x$

Question 8. It is given that $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 8x$.

a) Show that a particular integral of the form $y = ax + b$, where a and b are constants, is unsuitable.

b) Show instead that a particular integral of the form $y = ax^2 + bx + c$, where a, b and c are constants, is suitable.

c) Hence find the general solution of the differential equation.

Particular Integrals of the form $y = ax^2 + bx + c$ where a, b and c are constants

Question 9. Find the values of a, b and c such that $y = ax^2 + bx + c$ is a solution of the differential equation $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 12x^2 + 6x - 3$.

Question 10. Find a particular integral for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} - 9y = 18x^2 + 11x - 14$

b) $4\frac{d^2y}{dx^2} - 16y = -8x^2$

c) $9\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 11 - 4x - 10x^2$

Question 11. Using your answers to question 10, find the general solution for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} - 9y = 18x^2 + 11x - 14$

b) $4\frac{d^2y}{dx^2} - 16y = -8x^2$

c) $9\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 11 - 4x - 10x^2$

Mix

Question 12. Find the general solution to each of the following differential equations.

a) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 4x + 1$

b) $\ddot{x} - 4\dot{x} + 4x = 4t^2 + 8t + 6$

c) $\frac{d^2y}{dx^2} + 9y = 9x$

d) $\ddot{x} - 5\dot{x} + 6x = 12$

e) $\frac{d^2y}{dx^2} + 4y = 8x^2$

f) $\ddot{x} + 4\dot{x} + 8x = 2t^2 + 4t$

g) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 4x + 2$

h) $25\ddot{x} + 10\dot{x} + x = 6 - t^2$

Question 13. Solve the differential equation $\frac{d^2y}{dx^2} + 25y = 50x^2 + 25x$ given that $y = 0$ when $x = 0$, and $\frac{dy}{dx} = 1$ when $x = 0$.

Question 14. Solve the differential equation $4\ddot{x} - 4\dot{x} + x = 4t + 5$ given that $x = 1$ when $t = 0$, and $\dot{x} = 0$ when $t = 0$.

Particular Integrals of the form $y = ke^{px}$ where k and p are constants

Question 15. Find the value of k such that $y = ke^x$ is a solution of the differential equation, where k is a constant: $\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 14y = 12e^x$.

Question 16. Find the value of k such that $y = ke^{2x}$ is a solution of the differential equation, where k is a constant: $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 12y = 8e^{2x}$.

Question 17. Find a particular integral for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 5e^{4x}$

b) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 3e^{-2x}$

c) $2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 4e^{x}$

Question 18. Using your answers to question 17, find the general solution for each of the following differential equations.

a) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 5e^{4x}$

b) $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 3e^{-2x}$

c) $2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 4e^{x}$

Question 19.

a) Solve the differential equation $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 0$. It is given that $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 6e^{3x}$.

b) Substitute $y = ke^{3x}$ into the differential equation and explain why this trial function fails.

c) Find the value of k such that $y = kxe^{3x}$ is a solution of the differential equation.

d) Write down the general solution of the differential equation.

Question 20. Find the general solution to each of the following differential equations.

a) $2\ddot{x} - 5\dot{x} - 3x = 6e^{3t}$

b) $4\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = 5e^{x}$

c) $4\ddot{x} + 81x = 8e^{\frac{1}{2}t}$

d) $9\frac{d^2y}{dx^2} - 24\frac{dy}{dx} + 52y = 17e^{-x}$

Question 21. Solve the differential equation $\ddot{x} + 9x = 9e^{-2t}$ given that $x = 0$ when $t = 0$, and $\dot{x} = 9$ when $t = 0$.

Question 22. It is given that $\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} - 12y = 60e^{-3x}$. Given also that $y \to 0$ as $x \to \infty$, and $\dfrac{dy}{dx} = 0$ when $x = 0$, solve the differential equation.

Question 23.

a) Solve the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$. It is given that $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 5e^{2x}$.

b) Substitute $y = ke^{2x}$ into the differential equation and explain why this trial function fails.

c) Substitute $y = kxe^{2x}$ into the differential equation and explain why this trial function fails.

d) Find the value of k such that $y = kx^2e^{2x}$ is a solution of the differential equation.

e) Write down the general solution of the differential equation.

Particular Integrals of the form $y = p\cos kx + q\sin kx$ where k, p and q are constants

Question 24. Find the values of the constants p and q such that $y = p\cos 3x + q\sin 3x$ is a solution of the differential equation $\frac{d^2y}{dx^2} + 10\frac{dy}{dx} + 9y = \sin 3x$.

Question 25. Find the values of the constants p and q such that $y = p\cos 4x + q\sin 4x$ is a solution of the differential equation $\dfrac{d^2y}{dx^2} + 9\dfrac{dy}{dx} - 36y = 5\cos 4x$.

Question 26. Find a particular integral for each of the following differential equations.

a) $25\dfrac{d^2y}{dx^2} - 9y = 9\cos 6x$

b) $2\dfrac{d^2y}{dx^2} - 11\dfrac{dy}{dx} + 5y = 89\sin 8x$

c) $3\dfrac{d^2y}{dx^2} + 11\dfrac{dy}{dx} + 6y = 20\sin 2x - 4\cos 2x$

Question 27. Using your answers to question 26, find the general solution for each of the following differential equations.

a) $25\dfrac{d^2y}{dx^2} - 9y = 9\cos 6x$

b) $2\dfrac{d^2y}{dx^2} - 11\dfrac{dy}{dx} + 5y = 89\sin 8x$

c) $3\dfrac{d^2y}{dx^2} + 11\dfrac{dy}{dx} + 6y = 20\sin 2x - 4\cos 2x$

Question 28.

a) Solve the differential equation $\dfrac{d^2y}{dx^2} + 4y = 0$. It is given that $\dfrac{d^2y}{dx^2} + 4y = \sin 2x$.

b) Substitute $y = p\cos 2x + q\sin 2x$ into the differential equation and explain why this trial function fails.

c) Find the values of p and q such that $y = px\cos 2x + qx\sin 2x$ is a solution of the differential equation.

d) Write down the general solution of the differential equation.

Mix

Question 29. It is given that $\frac{d^2y}{dx^2}+3\frac{dy}{dx}+2y=f(x)$, and the complementary function is $y=Ae^{-2x}+Be^{-x}$. Write down the trial function when:

a) $f(x)=5$
b) $f(x)=4e^{3x}$
c) $f(x)=2x+8$
d) $f(x)=3\sin x$
e) $f(x)=6x^2$
f) $f(x)=3e^{-x}$
g) $f(x)=8\cos 4x$
h) $f(x)=x$
i) $f(x)=5\cos 7x-9\sin 7x$
j) $f(x)=8e^{-2x}$

Question 30. It is given that $\frac{d^2y}{dx^2}+36y=f(x)$, and the complementary function is $y=A\cos 6x+B\sin 6x$. Write down the trial function when:

a) $f(x)=4e^{-x}$
b) $f(x)=1+2x$
c) $f(x)=3\cos 2x$
d) $f(x)=x^2-3$
e) $f(x)=6$
f) $f(x)=2\sin 6x$
g) $f(x)=7e^{6x}$
h) $f(x)=8\cos 6x$
i) $f(x)=2\sin 9x$
j) $f(x)=5\cos 6x-7\sin 6x$

Question 31. It is given that $\frac{d^2y}{dx^2} + 9\frac{dy}{dx} - 52y = f(x)$. Find the general solution for this differential equation when:

a) $f(x) = 12 - 13x$
b) $f(x) = 3e^{5x}$
c) $f(x) = 17e^{4x}$
d) $f(x) = 89\cos 3x + 89\sin 3x$

Question 32. It is given that $4\frac{d^2y}{dx^2} - 20\frac{dy}{dx} + 25y = f(x)$. Find the general solution for this differential equation when:

a) $f(x) = 841\cos x$
b) $f(x) = 18e^{x}$
c) $f(x) = 25 - 125x$
d) $f(x) = e^{\frac{5}{2}x}$

Prerequisite knowledge for Q33: Differentiation of Hyperbolic Functions

Question 33. It is given that $\ddot{x} + 2\dot{x} + 2x = f(t)$, where $x = 0$ when $t = 0$, and $\dot{x} = 0$ when $t = 0$. Find the particular solution to this differential equation when:

a) $f(t) = 2e^{-t}$
b) $f(t) = 5\sin t$
c) $f(t) = 6t^2$
d) $f(t) = 4\sinh t$
e) $f(t) = \cosh 2t - \sinh 2t$

Particular Integrals that are sums of polynomials, exponential functions or trigonometric functions

Question 34. Find the values of the constants p and q such that $y = pe^{2x} + qe^{4x}$ is a solution of the differential equation $5\dfrac{d^2y}{dx^2} + 9\dfrac{dy}{dx} - 2y = 4e^{2x} - 6e^{4x}$.

Question 35. Find the values of the constants p, q, r and k such that $y = p + qx + r\cos 2x + k\sin 2x$ is a solution of the differential equation $4\dfrac{d^2y}{dx^2} + y = 9x + \cos 2x$.

Question 36. Solve the differential equation $\dfrac{d^2y}{dx^2} - y = x + \sin x$.

Question 37. It is given that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 42y = f(x)$, and the complementary function is $y = Ae^{-6x} + Be^{7x}$. Write down the trial function when:

a) $f(x) = 3 + e^x$
b) $f(x) = 3e^{2x} - e^x$
c) $f(x) = 6x - e^{5x}$
d) $f(x) = e^{4x} + 2\sin x$
e) $f(x) = \cos 3x - x^2$
f) $f(x) = 5 + 3x + e^{-5x}$
g) $f(x) = x - \sin \frac{1}{2}x - \cos \frac{1}{2}x$
h) $f(x) = e^{-6x} - 8$
i) $f(x) = e^{6x} + 8e^{7x} + x$
j) $f(x) = 5e^{-6x} - e^{7x} + \cos 3x$

Question 38. It is given that $\frac{d^2y}{dx^2} - 16\frac{dy}{dx} + 64y = f(x)$, and the complementary function is $y = Axe^{8x} + Be^{8x}$. Write down the trial function when:

a) $f(x) = \sin 3x - 7$
b) $f(x) = 3x - 10e^{-8x}$
c) $f(x) = e^x - 6\cos \frac{3}{2}x$
d) $f(x) = 5e^{4x} - 8 - 4x$
e) $f(x) = 9\sin 12x - 8 - 3\cos 12x$
f) $f(x) = 11e^{-3x} - e^{-2x} + 4x^2$
g) $f(x) = \sin x - \cos 2x$
h) $f(x) = 5x^2 + 3x - 8 + e^{10x}$
i) $f(x) = e^{8x} - 7x^2$
j) $f(x) = 2\cos 9x - 4\sin 4x - x^2 + 5e^{-8x}$

Question 39. Solve the differential equation $\dfrac{d^2y}{dx^2} - 5\dfrac{dy}{dx} + 6y = 3e^{2x} + 2e^{3x}$.

Question 40. Solve the differential equation $25\dfrac{d^2y}{dx^2} - 40\dfrac{dy}{dx} + 16y = 4 - 9e^{-\frac{2}{5}x} - 10e^{\frac{4}{5}x}$.

Question 41. It is given that $\ddot{x} - 7\dot{x} + 6x = f(t)$, where $x = 0$ when $t = 0$, and $\dot{x} = 1$ when $t = 0$. Find the particular solution for this differential equation when:

a) $f(t) = 72t + 74\sin t$
b) $f(t) = 50e^{t} - 52e^{2t}$
c) $f(t) = 35e^{6t} - 36t$
d) $f(t) = 108t^{2} - 100\cos 2t$

Question 42. Solve the differential equation $\dfrac{d^2y}{dx^2} + y = \sin x + \sin 2x$ given that $y = 5$ when $x = 0$, and $\dfrac{dy}{dx} = 3$ when $x = 0$.

Question 43. It is given that $\dfrac{d^2y}{dx^2}+2\dfrac{dy}{dx}+5y=17\cos 2x+10\sin x$. A solution to the differential equation is shown in the diagram below. It is given that the curve has a maximum point at $(0,10)$. Find the equation of the curve.

